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Bihar Engineering University, Patna

B.Tech 1st Semester Exam-2022

Time: 03 Hours

Code: 100105 ***Subject: Workshop Manufacturing Practices*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following: (any seven) **[14]**

(a) Structural sections such as rails, angles, I-beams are made by
(i) Hot rolling
(ii) Hot drawing
(iii) Hot piercing
(iv) Hot extrusion

(b) In arc welding, the electric arc is produced between the work and the electrode by
(i) Voltage
(ii) Flow of current
(iii) Contact resistance
(iv) All of the above

(c) Blanking is a
(i) Forging operation
(ii) Sheet metal operation
(iii) Bending operation
(iv) Rolling operation

(d) Mallet is a
(i) Measuring tool
(ii) Marking tool
(iii) Holding tool
(iv) Striking tool

(e) The temperature at which the new grains are formed in the metal is called
(i) Eutectic temperature
(ii) Eutectoid temperature
(iii) Critical temperature
(iv) Recrystallization temperature

(f) In centrifugal casting
(i) Core is made of sand
(ii) Core is made of ferrous material
(iii) Core is made of non-ferrous material
(iv) No core is used

(g) Cutting and forming operations can be performed in a single operation in a
(i) Simple die
(ii) Progressive die
(iii) Compound die
(iv) Combination die

(h) Loam sand is a mixture of
(i) 20% sand and 80% clay
(ii) 50% sand and 50% clay
(iii) 70% sand and 30% clay
(iv) 80% sand and 20% clay

(i) Which of the following welding processes uses non-consumable electrodes?
(i) TIG welding
(ii) MIG welding
(iii) Manual arc welding
(iv) Submerged arc welding

(j) A ring gauge is used to
(i) Check the diameter of shafts or studs
(ii) Test the accuracy of holes
(iii) Check the clearance between two mating surfaces
(iv) All of the above

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Q.2 Solve this question : **[14]**

(a) List any six forging defects.
Distinguish between blanking and punching.
Explain any two stakes used in sheet metal work with sketches.

Q.3 Solve both questions : **[7+7=14]**

(a) List out the different types of files. Write the use of any three types of files with sketches.

(b) Describe the working of spring hammer with neat sketch.

Q.4 Solve both questions : **[8+6=14]**

(a) Write four advantages of hot working process. Explain the principle of hot rolling.

(b) Explain the following hot working processes: i. Extrusion ii. Drawing

Q.5 Solve both questions : **[8+6=14]**

(a) List out various fastening methods those are used in sheet metal and explain any two of them.

(b) What is the difference between hem and seam?

Q.6 Solve both questions : **[6+8=14]**

(a) Write the specific application of any three measuring tools in carpentry. List out the different holding tools used in carpentry.

(b) Explain the following handsaws used in carpentry with neat sketch: i. Rip saw ii. Bow saw iii. Dovetail Saw iv. Keyhole saw

Q.7 Solve both questions : **[7+7=14]**

(a) Discuss different type of electrodes and why electrode is coating.

(b) What are the welding defects occurs during welding. Explain any four.

Q.8 Solve all parts : **[14]**

(a) List any four materials used for pattern making.

(b) Explain any four properties of moulding sand.

(c) What is meant by sand binder? Name the types of sand binders.

Q.9 Solve all parts : **[14]**

(a) What is 3D printing? Write some important applications of 3D Printing.

(b) Discuss Fused Deposition Modeling process of 3D printing.

(c) Differentiate between Soldering and Brazing.

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